

Electrodynamics

A Treatise on Electricity and Magnetism

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My Website

Chapter 0

Preface

These are notes I've made for my courses on Electromagnetism (PHY209) that I am taking now at IISER Bhopal. I've mostly used the textbooks *Introduction to Electrodynamics* by D.J. Griffiths and *Modern Electrodynamics* by Andrew Zangwill as references to create these notes.

Some conventions and notation I'm following are the Einstein Summation Convention, which states that if an index is repeated exactly twice, then it is to be summed over. For example, I'd write the dot product as $\boldsymbol{A} \cdot \boldsymbol{B} = A_i B_i$ while ommitting the summation symbol. I use the regular $d\boldsymbol{a}$ that Griffiths uses, instead of something like $d\boldsymbol{S}$ for area infinitesimals, but I replace $\int_{\mathcal{S}}$ with $\iint_{\mathcal{S}}$ to better imply that its a double integral that being done.

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Chapter 1

Preliminaries

"The enormous usefulness of mathematics in the natural sciences is something bordering on the mysterious..."

"The unreasonable effectiveness of mathematics in the natural sciences", Communications on Pure and Applied Mathematics
Eugene Wigner

1.1 Grad, Div and Curl

We talk about importance of the gradient, the divergence and the curl, and how one applied them to scalar and vector fields.

1.1.1 Scalar and Vector Fields

A field is an object that prescribes a different object to every point in space. If the prescribed is a scalar, then it is a scalar field and takes the form of a function of (usually) three variables $\phi(x, y, z)$. If the prescribed object is a vector, then it is a vector field denoted by $\boldsymbol{v}(x, y, z)$. Each component of a vector field is an individual scalar field in the each direction.

$$\boldsymbol{v}(x, y, z) = v_x(x, y, z)\hat{\boldsymbol{x}} + v_y(x, y, z)\hat{\boldsymbol{y}} + v_z(x, y, z)\hat{\boldsymbol{z}} \tag{1.1}$$

1.1.2 Gradient

Definition 1.1 – Nabla Operator

One defines the Nabla/Del operator to be the following:

$$\nabla = \hat{\boldsymbol{x}}\frac{\partial}{\partial x} + \hat{\boldsymbol{y}}\frac{\partial}{\partial y} + \hat{\boldsymbol{z}}\frac{\partial}{\partial z} \tag{1.2}$$

We shall discuss how one applies this to scalar fields $\phi(x, y, z)$. The vector $\nabla\phi(\boldsymbol{r})$ is called the gradient of the scalar field ϕ at a point $\boldsymbol{r} = (x, y, z)$.

$$\nabla\phi = \frac{\partial\phi}{\partial x}\hat{\boldsymbol{x}} + \frac{\partial\phi}{\partial y}\hat{\boldsymbol{y}} + \frac{\partial\phi}{\partial z}\hat{\boldsymbol{z}} \tag{1.3}$$

To interpret what this means, we make use of the multivariate chain rule to write the differential of ϕ as:

$$d\phi = \frac{\partial\phi}{\partial x}dx + \frac{\partial\phi}{\partial y}dy + \frac{\partial\phi}{\partial z}dz \tag{1.4}$$

If we define a displacement vector $d\boldsymbol{l} = dx\hat{\boldsymbol{x}}+dy\hat{\boldsymbol{y}}+dz\hat{\boldsymbol{z}}$, then we can rewrite the differential of ϕ as:

$$d\phi = \nabla\phi \cdot d\boldsymbol{l} = |\nabla\phi||d\boldsymbol{l}|\cos\theta \tag{1.5}$$

If we fix the magnitude of $d\boldsymbol{l}$, then it turns out the maximum change in ϕ happens when the angle between $d\boldsymbol{l}$ and $\nabla\phi$ is zero, i.e. $\cos\theta = 1$. Therefore, we say that *the gradient points in the direction of maximum change, and its magnitude gives us the rate of increase in this maximal direction.*

If the gradient of a scalar field vanishes, then that point is a stationary point which could be a maximum (a summit), a minimum (a valley), a saddle point (a pass), or a "shoulder.

1.1.3 Divergence

The dot product of the nabla operator with a vector field $\boldsymbol{v}$ is called the divergence of that vector field.

$$\nabla \cdot \boldsymbol{v} = \frac{\partial v_x}{\partial x} + \frac{\partial v_y}{\partial y} + \frac{\partial v_z}{\partial z} \equiv \frac{\partial v_i}{\partial x_i} \tag{1.6}$$

The divergence of a vector about a point $\boldsymbol{r}$ gives the value of the "spread" of the vector field. One way to imagine it is by considering the flow of water. If you drop some water and you see it spread out or diverge, then the divergence there is positive, since there is a spread of water. Using this, we define the terms "faucet" or "source" for a point with positive divergence and "sink" or "drain" for negative divergence. If a point has zero divergence, there is a equal amount of source and sink there, or no flow at all.

1.1.4 Curl

The cross product operation of nabla on a vector field is called the curl of the vector field,

$$(\nabla \times \boldsymbol{v})_i = \varepsilon_{ijk}\frac{\partial v_j}{\partial x_k} \tag{1.7}$$

where the index go in the order x, y, z and x_k is the k coordinate. It is reccomended to do cross products using the Levi–Civita symbol ε_{ijk} (please do this, it made my life doing cross product so much easier).

Definition 1.2 – Levi–Civita and Kronecker Delta

The Levi–Civita and Kronecker Delta symbols are tensors which are defined as follows:

$$\varepsilon_{ijk} = \begin{cases} 1, & \text{cyclic permutation } (xyz, yzx, zxy) \\ -1, & \text{anticyclic permutation } (zyx, yxz, xzy) \\ 0, & \text{otherwise} \end{cases} \tag{1.8}$$

$$\delta_{nm} = \begin{cases} 0, & m \neq n \\ 1, & m = n \end{cases} \tag{1.9}$$

One should also note the identity:

$$\varepsilon_{ijk}\varepsilon_{klm} = \delta_{il}\delta_{jm} - \delta_{im}\delta_{jl} \tag{1.10}$$

The curl is a measure of how much and what direction the vector $\boldsymbol{v}$ swirls around the point in question. For example, a voxtex or a whirlpool is an example of a large magnitude of curl.

1.2 Integrals

1.2.1 Line Integrals

A line integral is a way of integrating a vector valued function along a specific curve/path in space. If we wish to integrate over a path $\mathcal{P}$, the line integral is written as:

$$\int_{\mathcal{P}} \boldsymbol{v} \cdot d\boldsymbol{l}$$

If the path is a closed path, we write it as:

$$\oint_{\mathcal{P}} \boldsymbol{v} \cdot d\boldsymbol{l}$$

Essentially, we are taking the sum of components of the vector field that lies along this path $\mathcal{P}$. If we have a vector field that is independent of the path taken, then it is called a *conservative field*.

To illlustrate how one calculates a line integral, consider the following example:

Example 1.1 – Work done by a Force

Calculate the work done by the force $\boldsymbol{F} = y^2\hat{\boldsymbol{x}} + 2x\hat{\boldsymbol{y}}$ Newton in moving a body from $(1, -1)$ to $(1, 1)$ over a parabolic path of $y = x^2$.
Solution: First, we know the work done by a force over a path is defined as the

line integral of the force over that path.

$$W = \int_{\mathcal{P}} \boldsymbol{F} \cdot d\boldsymbol{l} = \int_{\mathcal{P}} (y^2 \hat{\boldsymbol{x}} + 2x\hat{\boldsymbol{y}}) \cdot (dx\, \hat{\boldsymbol{x}} + dy\, \hat{\boldsymbol{y}}) \tag{1.11}$$

We now try to rewrite this in terms of x, by using the fact that $y = x^2 \implies dy = 2x\, dx$.

$$W = \int_{(-1,1)}^{(1,1)} y^2\, dx + 2x\, dy = \int_{-1}^{1} x^4 + 4x^2\, dx = \frac{46}{15} \text{ Joule} \tag{1.12}$$

1.2.2 Surface Integrals

Surface integrals are the sum of the component of the vector field lying in the direction of the surface area, which we take to be outward by convention. It represents what portion of the field actually passes through the surface $\mathcal{S}$.

$$\iint_{\mathcal{S}} \boldsymbol{v} \cdot d\boldsymbol{a}$$

If we have a closed surface, we use the following symbol:

$$\oiint_{\mathcal{S}} \boldsymbol{v} \cdot d\boldsymbol{a}$$

Example 1.2 – Electric Flux through a Surface

Consider a unit cube in the first octant with one of its corners at the origin, and an electric field $\boldsymbol{E} = 2xz\, \hat{\boldsymbol{x}} + 2y\, \hat{\boldsymbol{y}} + (xy + z)\, \hat{\boldsymbol{z}}$, now find the total flux through the cube.

Solution: The electric flux through a surface $\mathcal{S}$ is defined to be:

$$\Phi_E = \iint_{\mathcal{S}} \boldsymbol{E} \cdot d\boldsymbol{a} \tag{1.13}$$

A cube has six sides, and taking one side at a time:

1. $x = 1$, $d\boldsymbol{a} = dy\, dz\, \hat{\boldsymbol{x}}$

$$\Phi_E = \int_0^1 \int_0^1 2z\, dy\, dz = 1 \tag{1.14}$$

2. $x = 0$, $d\boldsymbol{a} = -dy\, dz\, \hat{\boldsymbol{x}}$

$$\Phi_E = \int_0^1 \int_0^1 0\, dy\, dz = 0 \tag{1.15}$$

3. $y = 1$, $d\boldsymbol{a} = dx\, dz\, \hat{\boldsymbol{y}}$

$$\Phi_E = \int_0^1 \int_0^1 2\, dz\, dx = 2 \tag{1.16}$$

4. $y = 0$, $d\boldsymbol{a} = -dx\, dz\, \hat{\boldsymbol{y}}$

$$\Phi_E = \int_0^1 \int_0^1 0\, dx\, dz = 0 \tag{1.17}$$

5. $z = 1$, $d\boldsymbol{a} = dx\,dy\,\hat{\boldsymbol{x}}$

$$\Phi_E = \int_0^1 \int_0^1 xy + 1\,dx\,dy = 1.25 \tag{1.18}$$

6. $z = 0$, $d\boldsymbol{a} = -dx\,dy\,\hat{\boldsymbol{z}}$

$$\Phi_E = \int_0^1 \int_0^1 -xy\,dx\,dy = -0.25 \tag{1.19}$$

The total flux is the sum of all these, which is $\Phi_E = 4\ \mathrm{Nm^2C^{-1}}$

1.2.3 Volume Integrals

A volume integral is simply a sum of a quantity, for example density, over a volume to give mass. It is expressed as:

$$\iiint_{\mathcal{V}} \rho\,d\tau \tag{1.20}$$

where the cartesian volume element $d\tau = dx\,dy\,dz$. You can follow the given example to understand them better:

Example 1.3 – Mass of a Cube

A unit cube in the first octant with one of its corners at the origin has its density given by $\rho(x, y, z) = x^2 + 2zy$ kg m^{-3}, find its mass.

Solution: The mass is a simple volume integral of the density.

$$m = \iiint_{\mathcal{V}} \rho(x, y, z)\,d\tau = \int_0^1 \int_0^1 \int_0^1 x^2 + 2yz\,dx\,dy\,dz \tag{1.21}$$

$$= \int_0^1 \int_0^1 \frac{x^3}{3} + 2xyz\Big|_0^1 dy\,dz = \int_0^1 \int_0^1 \frac{1}{3} + 3yz\,dy\,dz \tag{1.22}$$

$$= \int_0^1 \frac{y}{3} + y^2 z\Big|_0^1 dz = \int_0^1 \frac{1}{3} + z\,dz \tag{1.23}$$

$$= \frac{z}{3} + \frac{z^2}{2}\Big|_0^1 = \frac{5}{6}\ \mathrm{kg} \tag{1.24}$$

1.3 Integral and Nabla Theorems

1.3.1 Gradient Theorem

Let $f(x, y, z)$ be a scalar function, and let $\boldsymbol{F} = \nabla f$. Now, we attempt to take the line integral of $\boldsymbol{F}$ over any path $\mathcal{P}$ from points $\boldsymbol{a}$ to $\boldsymbol{b}$.

$$\int_{\boldsymbol{a}}^{\boldsymbol{b}} \boldsymbol{F} \cdot d\boldsymbol{l} = \int_{\boldsymbol{a}}^{\boldsymbol{b}} \nabla f \cdot d\boldsymbol{l} = \int_{\boldsymbol{a}}^{\boldsymbol{b}} \underbrace{\frac{\partial f}{\partial x}dx + \frac{\partial f}{\partial y}dy + \frac{\partial f}{\partial z}dz}_{df} \tag{1.25}$$

$$= \int_{\boldsymbol{a}}^{\boldsymbol{b}} df(x, y, z) = f(\boldsymbol{b}) - f(\boldsymbol{a}) \tag{1.26}$$

Essentially, we just proved that the line integral of the gradient of a function is always independent of the path chosen between two points. The function $f(x, y, z)$ is called the *scalar potential* of $\boldsymbol{F}$, and $\boldsymbol{F}$ is called a *conservative field*.

1.3.2 Divergence Theorem

Consider a volume $\mathcal{V}$, which has a closed surface $\mathcal{S}$, then integral of a vector field $\boldsymbol{v}$ over that surface is equal to the volume integral of the divergence of $\boldsymbol{v}$.

$$\oiint_{\mathcal{S}} \boldsymbol{v} \cdot d\boldsymbol{a} = \iiint_{\mathcal{V}} \nabla \cdot \boldsymbol{v}\, d\tau \tag{1.27}$$

The formal proof for it is a bit difficult, so we shall not cover it. An interpretation of this theorem is if we consider the sources inside the volume, the surface integral gives us the actual flow that enters/exits, which should equate to the volume integral that gives us the sum of the spread of the flow.

1.3.3 Stokes' Theorem

Take an open surface $\mathcal{S}$ which has a boundary, a closed path $\mathcal{P}$. Then, we have the following:

$$\oint_{\mathcal{P}} \boldsymbol{v} \cdot d\boldsymbol{l} = \iint_{\mathcal{S}} (\nabla \times \boldsymbol{v}) \cdot d\boldsymbol{a} \tag{1.28}$$

This is called *Stokes' Theorem*, and one may interpret this as the sum of the curls inside the surface is equal to the big curl around the boundary of that surface. For conservative forces, one can note that both sides are indeed zero, because the path integral is a difference of function at endpoints, and curl of a gradient is zero.

1.4 Coordinate Transformations

1.4.1 Jacobian Matrix

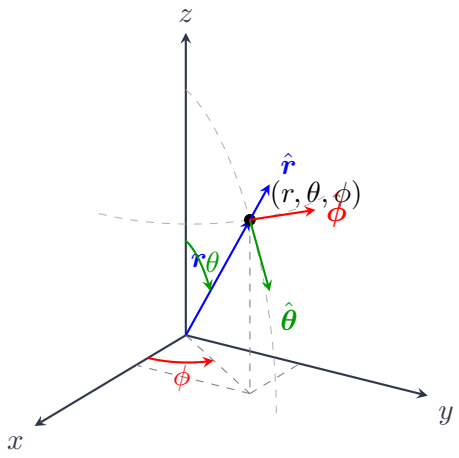
Let us say we wish transform one coordinate system, denoted by $\boldsymbol{x}$ into another, denoted by $\boldsymbol{y}$. If we have to perform volume integrals, we will have different volume elements for each system, but we can relate them using the *Jacobian matrix*.

$$\boldsymbol{J}(\boldsymbol{x},\boldsymbol{y}) = \begin{pmatrix} \frac{\partial x_1}{\partial y_1} & \frac{\partial x_1}{\partial y_2} & \cdots & \frac{\partial x_1}{\partial y_n} \\ \frac{\partial x_2}{\partial y_1} & \frac{\partial x_2}{\partial y_2} & \cdots & \frac{\partial x_2}{\partial y_n} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial x_n}{\partial y_1} & \frac{\partial x_n}{\partial y_2} & \cdots & \frac{\partial x_n}{\partial y_n} \end{pmatrix} \tag{1.29}$$

The actual relation is given by the Jacobian determinant, which is often also denoted as $|\partial\boldsymbol{x}/\partial\boldsymbol{y}|$.

$$d\tau_{\boldsymbol{x}} = |\boldsymbol{J}(\boldsymbol{x},\boldsymbol{y})|\, d\tau_{\boldsymbol{y}} \tag{1.30}$$

1.4.2 Spherical Coordinates



Let (x, y, z) be the position of a point in cartesian coordinates. Then we define (r, θ, ϕ) to be the position using spherical coordinates. r is the distance of the point from the origin, θ is the angle made with the $+z$–axis, and ϕ is the angle made by the projection on the x–y plane of the line joining the point and origin with the x–axis. Using the given diagram, one can discern that:

$$x = r\sin\theta\cos\phi \tag{1.31}$$

$$y = r\sin\theta\sin\phi \tag{1.32}$$

$$z = r\cos\phi \tag{1.33}$$

And the cartesian to curvilinear transformation being:

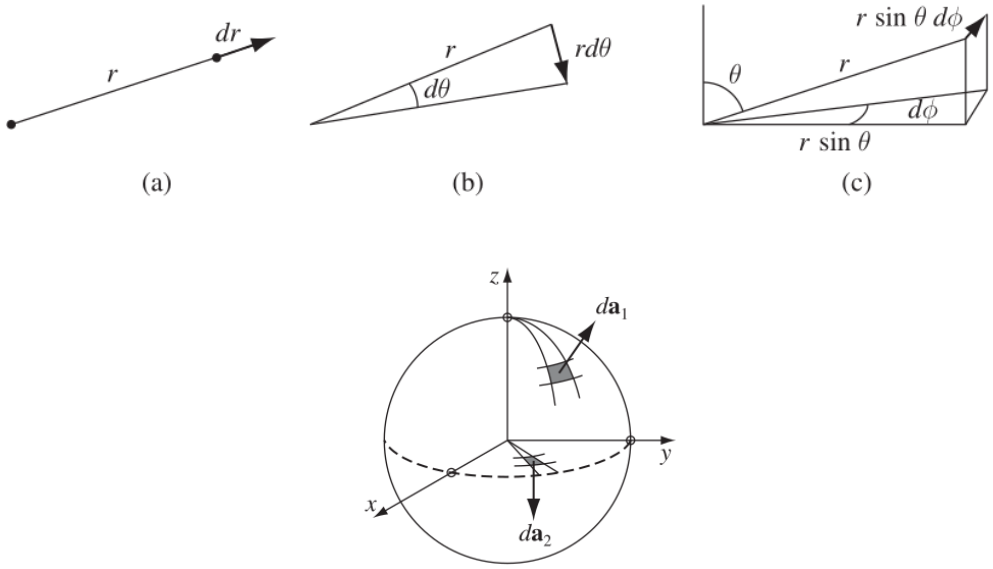
$$r = \sqrt{x^2+y^2+z^2} \in [0,\infty) \tag{1.34}$$

$$\theta = \tan^{-1}\left(\frac{\sqrt{x^2+y^2}}{z}\right) \in (0,\pi) \tag{1.35}$$

$$\phi = \tan^{-1}\left(\frac{y}{x}\right) \in (0,2\pi) \tag{1.36}$$

The unit vectors for the curvilinear coordinates are not fixed unlike the cartesian ones. This is a very important point to note, as the unit vectors will always be relative to the original point in space you are considering.

Now let's try to find the infinitesimal displacement vector $d\boldsymbol{l}$ for spherical coordinates. We do this by performing an infinitesimal transformation $(r, \theta, \phi) \to (r + dr, \theta + d\theta, \phi + d\phi)$



and then finding the displacement between the original and translated vectors. One can see that the radial part only increases in the radial direction, so $dl_r = dr$, but looking at the change in the $\hat{\boldsymbol{\theta}}$ direction, we see a displacement of $rd\theta$. Similarly, look at the $\hat{\boldsymbol{\phi}}$ direction, and we have a displacement of $r\sin\theta d\phi$. Therefore, we have the displacement vector to be:

$$d\boldsymbol{l} = dr\hat{\boldsymbol{r}} + rd\theta\hat{\boldsymbol{\theta}} + r\sin\theta\, d\phi\hat{\boldsymbol{\phi}} \tag{1.37}$$

We do not take it to be $d\boldsymbol{l} = dr\hat{\boldsymbol{r}} + d\theta\hat{\boldsymbol{\theta}} + d\phi\hat{\boldsymbol{\phi}}$ because this implies $d\theta$ and $d\phi$ have the units of length, which is simply not true since they are angles.

Another important quantity we consider is the infinitesimal volume area elements, $d\tau$ and $d\boldsymbol{a}$. The volume element is pretty simple, it is just the product of the components of the displacement vector, since it is small enough to be considered as a cuboid.

$$d\tau = dl_r dl_\theta dl_\phi = r^2\sin\theta\, dr\, d\theta\, d\phi \tag{1.38}$$

The general formula depends on the surface, but we know the answer for two cases: one is if the area we are taking is a part of a sphere, then like the volume case, we consider it to small enough to take it as a rectangle of length and breadth dl_θ and dl_ϕ, with the normal in the $\hat{\boldsymbol{r}}$ direction:

$$d\boldsymbol{a}_1 = dl_\theta dl_\phi\hat{\boldsymbol{r}} = r^2\sin\theta\, d\theta\, d\phi\,\hat{\boldsymbol{r}} \tag{1.39}$$

The second case is that of an area that lies on the x–y plane, and using the same technique, we see its unit normal is the $\hat{\boldsymbol{\theta}}$ direction and the vector to be:

$$d\boldsymbol{a}_2 = dl_r dl_\phi\hat{\boldsymbol{\theta}} = r\, dr\, d\phi\,\hat{\boldsymbol{\theta}} \tag{1.40}$$

To obtain the nabla/del operations in spherical coordinates, you can just use the multi-

variable chain rule:

$$\frac{\partial f}{\partial x} = \frac{\partial f}{\partial r}\frac{\partial r}{\partial x} + \frac{\partial f}{\partial \theta}\frac{\partial \theta}{\partial x} + \frac{\partial f}{\partial \phi}\frac{\partial \phi}{\partial x} \tag{1.41}$$

and we substitute it into the operators to obtain:

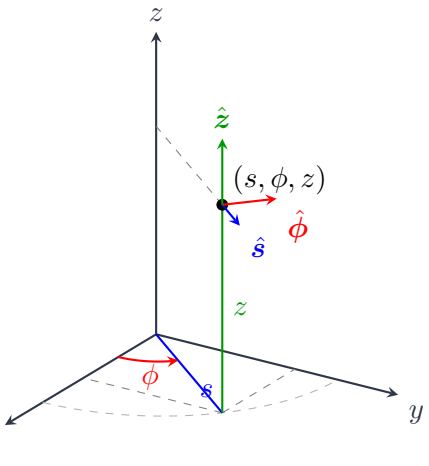
$$\nabla f = \frac{\partial f}{\partial r}\hat{\boldsymbol{r}} + \frac{1}{r}\frac{\partial f}{\partial \theta}\hat{\boldsymbol{\theta}} + \frac{1}{r\sin\theta}\frac{\partial f}{\partial \phi}\hat{\boldsymbol{\phi}} \tag{1.42}$$

$$\nabla \cdot \boldsymbol{v} = \frac{1}{r^2}\frac{\partial}{\partial r}(r^2 v_r) + \frac{1}{r\sin\theta}\frac{\partial}{\partial \theta}(\sin\theta\, v_\theta) + \frac{1}{r\sin\theta}\frac{\partial v_\phi}{\partial \phi} \tag{1.43}$$

$$\nabla \times \boldsymbol{v} = \frac{1}{r\sin\theta}\left[\frac{\partial}{\partial\theta}(\sin\theta\, v_\phi) - \frac{\partial v_\theta}{\partial \phi}\right]\hat{\boldsymbol{r}} + \frac{1}{r}\left[\frac{1}{\sin\theta}\frac{\partial v_r}{\partial \phi} - \frac{\partial}{\partial r}(r v_\phi)\right]\hat{\boldsymbol{\theta}} + \frac{1}{r}\left[\frac{\partial}{\partial r}(r v_\theta) - \frac{\partial v_r}{\partial \theta}\right]\hat{\boldsymbol{\phi}} \tag{1.44}$$

1.4.3 Cylindrical Coordinates

Let (x, y, z) represent a point in cartesian space, then the coordinates (s, ϕ, z) are the cylindrical coordinates, where s is the length of the projection of line joining the point and origin on the x–y plane, ϕ is the angle this projection makes with the x–axis (same as spherical) and z is the distance of the point from x–y plane (same as cartesian).



$$x = s\cos\phi \tag{1.45}$$

$$y = s\sin\phi \tag{1.46}$$

$$z = z \tag{1.47}$$

And the cartesian to cylindrical transformation is given by:

$$s = \sqrt{x^2 + y^2} \in [0, \infty) \tag{1.48}$$

$$\phi = \tan^{-1}\left(\frac{y}{x}\right) \in (-\pi, \pi) \tag{1.49}$$

$$z = z \in (-\infty, \infty) \tag{1.50}$$

In this case, we have a fixed $\hat{\boldsymbol{z}}$ vector and variable $\hat{\boldsymbol{s}}$ and $\hat{\boldsymbol{\phi}}$ vectors. Now for a line element/infinitesimal displacement $d\boldsymbol{l}$, we have it to be:

$$d\boldsymbol{l} = ds\,\hat{\boldsymbol{s}} + s d\phi\,\hat{\boldsymbol{\phi}} + dz\,\hat{\boldsymbol{z}} \tag{1.51}$$

and the volume element is:

$$d\tau = s\, ds\, d\phi\, dz \tag{1.52}$$

The nabla/del operations in cylindrical coordinates is given as:

$$\nabla f = \frac{\partial f}{\partial s}\hat{\boldsymbol{s}} + \frac{1}{s}\frac{\partial f}{\partial \phi}\hat{\boldsymbol{\phi}} + \frac{\partial f}{\partial z}\hat{\boldsymbol{z}} \tag{1.53}$$

$$\nabla \cdot \boldsymbol{v} = \frac{1}{s}\frac{\partial}{\partial s}(sv_s) + \frac{1}{s}\frac{\partial v_\phi}{\partial \phi} + \frac{\partial v_z}{\partial z} \tag{1.54}$$

$$\nabla \times \boldsymbol{v} = \left(\frac{1}{s}\frac{\partial v_z}{\partial \phi} - \frac{\partial v_\phi}{\partial z}\right)\hat{\boldsymbol{s}} + \left(\frac{\partial v_s}{\partial z} - \frac{\partial v_z}{\partial s}\right)\hat{\boldsymbol{\phi}} + \frac{1}{s}\left(\frac{\partial}{\partial s}(sv_\phi) - \frac{\partial v_s}{\partial \phi}\right)\hat{\boldsymbol{z}} \tag{1.55}$$

1.4.4 Spherical to & from Cylindrical

The conversion between spherical to cylindrical and cylindrical to spherical is given below:

$$s = r\sin\theta \iff r = \sqrt{s^2 + z^2} \tag{1.56}$$

$$\phi = \phi \iff \theta = \tan^{-1}\left(\frac{s}{z}\right) \tag{1.57}$$

$$z = r\cos\theta \iff \phi = \phi \tag{1.58}$$

The infinitesimals are given as:

$$\begin{pmatrix} ds \\ d\phi \\ dz \end{pmatrix} = \begin{pmatrix} \sin\theta & r\cos\theta & 0 \\ 0 & 0 & 1 \\ \cos\theta & -r\sin\theta & 0 \end{pmatrix} \begin{pmatrix} dr \\ d\theta \\ d\phi \end{pmatrix} \tag{1.59}$$

1.5 Dirac Delta Function

The Dirac Delta function is a particularly useful "function", which is defined as follows:

$$\delta(x) = \begin{cases} 0, & x \neq 0 \\ \infty, & x = 0 \end{cases} \tag{1.60}$$

It can be interpreted as a large spike at $x = 0$, and as a limit of rectangle of unit area. This unit area part is important because we also define, for any $\varepsilon > 0$:

$$\int_{-\varepsilon}^{\varepsilon} \delta(x)\,dx = 1 \tag{1.61}$$

Any integral that contains the origin will give an area of 1. Since $\delta(x)$ is zero everywhere except the origin, then $f(x)\delta(x) = f(0)\delta(x)$, or for a shifted delta, we have $f(x)\delta(x-a) = f(a)\delta(x-a)$.

$$\int_{a-\varepsilon}^{a+\varepsilon} f(x)\delta(x-a)\,dx = f(a) \tag{1.62}$$

We similarly define a 3D version of it, which we write as:

$$\delta^3(\boldsymbol{z}) = \delta(x-x')\delta(y-y')\delta(z-z') \tag{1.63}$$

Here, $\boldsymbol{\imath} = \boldsymbol{r} - \boldsymbol{r}'$, where $\boldsymbol{r} = (x, y, z)$ and $\boldsymbol{r}' = (x', y', z')$. And now generalizing equation 1.58, we have:

$$\iiint_{\boldsymbol{r}'-\varepsilon}^{\boldsymbol{r}'+\varepsilon} f(\boldsymbol{r})\delta^3(\boldsymbol{\imath})\,dx\,dy\,dz = f(\boldsymbol{r}') \tag{1.64}$$

Using the properties of the delta function, we find the gradient and divergence of the following fields as follows:

$$\nabla \cdot \left(\frac{\hat{\boldsymbol{\imath}}}{\imath^2}\right) = 4\pi\delta^3(\boldsymbol{\imath}) \tag{1.65}$$

$$\nabla^2 \left(\frac{1}{\imath}\right) = -4\pi\delta^3(\boldsymbol{\imath}) \tag{1.66}$$

Another important property of the Dirac Delta can be derived using Fourier and inverse Fourier transforms, which gives us:

$$\delta(x) = \frac{1}{2\pi}\int_{-\infty}^{+\infty} e^{ikx}\,dk \tag{1.67}$$

Another property, is that when transforming between coordinate systems, the delta function also transforms as follows:

$$\delta(\boldsymbol{x} - \boldsymbol{x}') = \frac{1}{|\boldsymbol{J}(\boldsymbol{x}, \boldsymbol{y})|}\delta(\boldsymbol{y} - \boldsymbol{y}') \tag{1.68}$$

1.6 Helmholtz Theorem

Given any arbitrary vector field $\boldsymbol{C}$, it can always be decomposed into a sum of two vector fields, one with zero divergence ($\boldsymbol{C}_1$) and the other with zero curl ($\boldsymbol{C}_2$). This implies that any field is uniquely determined by its curl and divergence.

$$\boldsymbol{C} = \boldsymbol{C}_1 + \boldsymbol{C}_2 \tag{1.69}$$

$$\nabla \cdot \boldsymbol{C}_1 = 0 \tag{1.70}$$

$$\nabla \times \boldsymbol{C}_2 = 0 \tag{1.71}$$

An particular representation of this that one may be interested is:

$$\boldsymbol{C} = \nabla \times \boldsymbol{A} - \nabla\varphi \tag{1.72}$$

where $\boldsymbol{A}$ and φ are given by:

$$\boldsymbol{A}(\boldsymbol{r}) = \frac{1}{4\pi}\iiint \frac{\nabla' \times \boldsymbol{C}(\boldsymbol{r}')}{\imath}\,d\tau' \tag{1.73}$$

$$\varphi(\boldsymbol{r}) = \frac{1}{4\pi}\iiint \frac{\nabla' \cdot \boldsymbol{C}(\boldsymbol{r}')}{\imath}\,d\tau' \tag{1.74}$$

Chapter 2

Electrostatics

"And we daily in our experiments electrise bodies plus or minus, as we think proper."

A Letter to Peter Collinson, 1747
Benjamin Franklin

2.1 Electric Fields

The forces between charged particles is, as we know, described by Coulomb's law:

$$\boldsymbol{F}_{qQ} = \frac{1}{4\pi\varepsilon_0}\frac{qQ}{r^2}\hat{\boldsymbol{r}} \tag{2.1}$$

where ε_0 is the *permittivity of free space* whose value is about $8.85\times10^{-12}\mathrm{C}^2/\mathrm{Nm}^2$. The force applied is inversely propotional to the square of the distance between the two charges. If we wish to examine the effect of a charge on any other charged partice, we turn to a format of analyzing fields instead of forces. So we define the electric field of a charge Q as follows:

$$\boldsymbol{E} = \frac{1}{4\pi\varepsilon_0}\frac{Q}{r^2}\hat{\boldsymbol{r}} = \frac{\boldsymbol{F}_{qQ}}{q} \tag{2.2}$$

The force on any n charges $q_1, q_2, \dots q_n$ due to the charge Q would be:

$$\boldsymbol{F} = \boldsymbol{F}_1 + \boldsymbol{F}_2 + \cdots + \boldsymbol{F}_n \tag{2.3}$$

$$= \frac{Q}{4\pi\varepsilon_0}\left(\frac{q_1}{r_1^2}\hat{\boldsymbol{r}}_1 + \frac{q_2}{r_2^2}\hat{\boldsymbol{r}}_2 + \cdots + \frac{q_n}{r_n^2}\hat{\boldsymbol{r}}_n\right) \tag{2.4}$$

$$\boldsymbol{F} = Q\boldsymbol{E} \tag{2.5}$$

2.1.1 Continuous Charge Distribution

Assume we have a continuous charge distribution described by either a linear density (λ), surface density (σ) or a volume density (ρ). For all of them, the electric field is given by

the following:

$$\boldsymbol{E}(\boldsymbol{r}) = \frac{1}{4\pi\varepsilon_0}\int \frac{1}{\boldsymbol{\imath}^2}\hat{\boldsymbol{\imath}}\, dq \tag{2.6}$$

$$\text{where } dq = \lambda\, dl' = \sigma\, da' = \rho\, d\tau'$$

Thus the electric field of a line charge is:

$$\boldsymbol{E}(\boldsymbol{r}) = \frac{1}{4\pi\varepsilon_0}\int_{\mathcal{P}} \frac{\lambda(\boldsymbol{r}')}{\boldsymbol{\imath}^2}\hat{\boldsymbol{\imath}}\, dl' \tag{2.7}$$

The electric field of a surface distributed charge:

$$\boldsymbol{E}(\boldsymbol{r}) = \frac{1}{4\pi\varepsilon_0}\iint_{\mathcal{S}} \frac{\sigma(\boldsymbol{r}')}{\boldsymbol{\imath}^2}\hat{\boldsymbol{\imath}}\, da' \tag{2.8}$$

And the electric field due a charge distributed over a volume is:

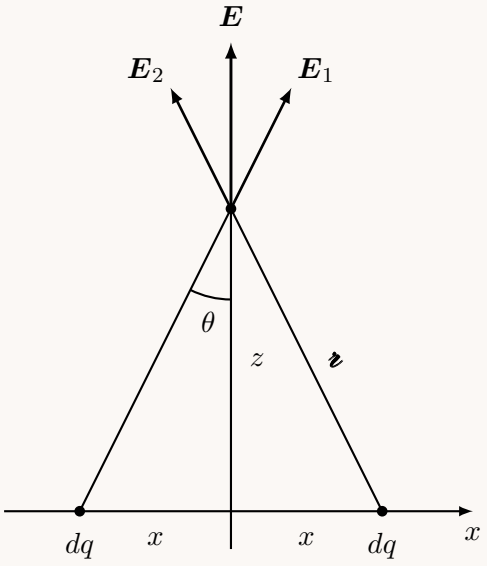
$$\boldsymbol{E}(\boldsymbol{r}) = \frac{1}{4\pi\varepsilon_0}\iiint_{\mathcal{V}} \frac{\rho(\boldsymbol{r}')}{\boldsymbol{\imath}^2}\hat{\boldsymbol{\imath}}\, d\tau' \tag{2.9}$$

We shall take an example of one to understand how one can calculate better.

Example 2.1 – Field due to a line segment distribution

ind the electric field of a of line segment of length $2L$ which carries a charge density of λ, at a distance z above its midpoint.

Solution: First, one should look for symmetries that one can exploit our problem easier. Here, if one takes two symmetrically placed points ($\pm x$), and the electric field of those two point charges should cancel out it the horizontal direction.



From this, one can infer that the electric field is:

$$d\boldsymbol{E} = 2\frac{1}{4\pi\varepsilon_0}\left(\frac{\lambda\, dx}{\imath^2}\right)\cos\theta\,\hat{\boldsymbol{z}} \tag{2.10}$$

Here $\cos\theta = z/\imath$ and $\imath = \sqrt{z^2+x^2}$ and x runs from 0 to L

$$E = \frac{1}{4\pi\varepsilon_0}\int_0^L \frac{2\lambda z}{(z^2+x^2)^{3/2}}\,dx \tag{2.11}$$

$$= \frac{\lambda z}{2\pi\varepsilon_0}\left[\frac{x}{z^2\sqrt{z^2+x^2}}\right]\Bigg|_0^L \tag{2.12}$$

$$= \frac{1}{4\pi\varepsilon_0}\frac{2\lambda L}{z\sqrt{z^2+L^2}} \tag{2.13}$$

If we have z far away then $z \gg L$, then the result simplifies to:

$$\boldsymbol{E}(z) = \frac{1}{4\pi\varepsilon_0}\frac{2\lambda L}{z^2}\hat{\boldsymbol{z}} \tag{2.14}$$

For the limit of an infinitely long wire i.e. $L \to \infty$ we get:

$$\boldsymbol{E}(z) = \frac{1}{4\pi\varepsilon_0}\frac{2\lambda}{z}\hat{\boldsymbol{z}} \tag{2.15}$$

2.1.2 Flux and Gauss' Law

Consider a surface $\mathcal{S}$ and an electric field. We define flux to be the "amount of electric field lines that pass through the surface", which is not really explanatory of what flux actually is, since it is a bit harder to visualise. Think of it as the amount of the field passing through an area. Since some of the field may not be directly aligned with the direction of the surface, we must only see the component in that direction, given by the dot product. Mathematically, flux is defined as:

$$\Phi_E = \iint_{\mathcal{S}} \boldsymbol{E}\cdot d\boldsymbol{a} \tag{2.16}$$

The dot product picks out the portion of the field lines that actually go through the normal of the surface at that point.

This suggests that if we consider a closed surface, then the charges inside are the ones most contributing most to it. This is the motivation for Gauss' law. Let us derive it quantitatively.
Consider a closed surface $\mathcal{S}$ and a charge q inside the volume enclosed by the surface. Now, we attempt to calculate the flux in spherical coordinates.

$$\Phi_E = \oiint_{\mathcal{S}} \boldsymbol{E}\cdot d\boldsymbol{a} = \oiint \frac{1}{4\pi\varepsilon_0}\frac{q}{r^2}\hat{\boldsymbol{r}}\cdot(r^2\sin\theta\, d\theta\, d\phi\,\hat{\boldsymbol{r}}) = \frac{q}{\varepsilon_0} \tag{2.17}$$

If we have a sum of charges then we replace $q = \sum_i q_i$ or if we have a charge density, then $q = \iiint_{\mathcal{V}} \rho \, d\tau$, where $\mathcal{V}$ is the volume enclosed by the surface. Essentially, this is all the charge enclosed inside the surface, and we obtain *Gauss' Law*:

$$\oiint_{\mathcal{S}} \boldsymbol{E} \cdot d\boldsymbol{a} = \frac{Q}{\varepsilon_0} \tag{2.18}$$

One may notice, that we can apply the divergence theorem here and take a charge density ρ inside the volume $\mathcal{V}$, we obtain:

$$\oiint_{\mathcal{S}} \boldsymbol{E} \cdot d\boldsymbol{a} = \iiint_{\mathcal{V}} (\nabla \cdot \boldsymbol{E}) \, d\tau = \frac{Q}{\varepsilon_0} \tag{2.19}$$

$$= \frac{1}{\varepsilon_0} \iiint_{\mathcal{V}} \rho(\boldsymbol{r}) \, d\tau \tag{2.20}$$

$$\nabla \cdot \boldsymbol{E} = \frac{\rho(\boldsymbol{r})}{\varepsilon_0} \tag{2.21}$$

This is the *Gauss' Law in differential form.* It is tidier, but typically the integral form accommodates point, line and surface charges better. This version is more useful for volume charge densities.

2.1.3 Divergence of $\boldsymbol{E}$

Let us try to once calculate the divergence of a general electric field, from the formula we derived in Equation (2.9).

$$\boldsymbol{E}(\boldsymbol{r}) = \frac{1}{4\pi\varepsilon_0} \iiint \rho(\boldsymbol{r}') \frac{\hat{\boldsymbol{\imath}}}{\imath^2} \, d\tau \tag{2.22}$$

$$\nabla \cdot \boldsymbol{E} = \frac{1}{4\pi\varepsilon_0} \iiint \rho(\boldsymbol{r}') \, \nabla \cdot \left(\frac{\hat{\boldsymbol{\imath}}}{\imath^2} \right) \, d\tau \tag{2.23}$$

If you remember from Equation (1.65) we calculated the divergence to be $\nabla \cdot (\hat{\boldsymbol{\imath}}/\imath^2) = 4\pi\delta^3(\boldsymbol{\imath})$.

$$\nabla \cdot \boldsymbol{E} = \frac{1}{4\pi\varepsilon_0} \iiint \rho(\boldsymbol{r}') \, 4\pi\delta^3(\boldsymbol{\imath}) \, d\tau \tag{2.24}$$

$$= \frac{1}{\varepsilon_0} \rho(\boldsymbol{r}) \tag{2.25}$$

Therefore, once finds that Gauss' law in differential form is much more universal than it seems. This is the first of the *Maxwell's equations* that we will be seeing.

Let's now take an example of applying Gauss' law to find the electric field due to a distribution.

Example 2.2 – Field due to hollow sphere

Consider a thick sphere of inner radius a and outer radius b with total charge Q, that is evenly spread. Find the electric field at a distance r from the centre of the sphere for the cases,

1. $r < a$
2. $a < r < b$
3. $r > b$

Solution.

1. Given we have a total charge of Q, then the charge density is:

$$\rho(r) = \frac{3Q}{4\pi(b^3 - a^3)}$$

   For the case $r < a$, we do not have any enclosed charge. By Gauss' law,

$$\oiint \boldsymbol{E} \cdot d\boldsymbol{a} = 0 \implies E(r) = 0, \quad (r < a)$$

2. For $a < r < b$, the charge countained within a sphere of radius r would be:

$$Q = \int_0^{2\pi} \int_0^{\pi} \int_a^r \rho(r')\,(r')^2 \sin\theta\, dr'\, d\theta\, d\phi = Q\left(\frac{r^3 - a^3}{b^3 - a^3}\right) \tag{2.26}$$

   Electric field and the normal to the surface have the same direction $\hat{\boldsymbol{r}}$, therefore:

$$\int_0^{2\pi} \int_0^{\pi} E\,\hat{\boldsymbol{r}} \cdot \hat{\boldsymbol{r}}\, r^2 \sin\theta\, d\theta\, d\phi = \frac{Q}{\varepsilon_0}\frac{r^3 - a^3}{b^3 - a^3} \implies E(r) = \frac{1}{4\pi\varepsilon_0}\frac{Q}{r^2}\frac{r^3 - a^3}{b^3 - a^3} \tag{2.27}$$

3. For $r > a$, we have a total enclosed charge of Q, then by applying Gauss' Law, we obtain:

$$E(r) = \frac{1}{4\pi\varepsilon_0}\frac{Q}{r^2} \tag{2.28}$$

2.1.4 Curl of $\boldsymbol{E}$

Again, from Coulomb's Law (2.9), we write the general electric field for a distribution as:

$$\boldsymbol{E}(\boldsymbol{r}) = \frac{1}{4\pi\varepsilon_0} \iiint_{\mathcal{V}} \frac{\rho(\boldsymbol{r}')}{\boldsymbol{\imath}^2}\hat{\boldsymbol{\imath}}\, d\tau' \tag{2.29}$$

We take the curl of this,

$$\nabla \times \boldsymbol{E} = \frac{1}{4\pi\varepsilon_0} \iiint_{\mathcal{V}} \rho(\boldsymbol{r}')\, \nabla \times \left(\frac{\hat{\boldsymbol{\imath}}}{\boldsymbol{\imath}^2}\right) d\tau' \tag{2.30}$$

It is easy to check that the curl of $\hat{\boldsymbol{\imath}}/\boldsymbol{\imath}^2$ is zero, and hence we obtain the curl of the electric field to be zero as well.

$$\nabla \times \boldsymbol{E} = 0 \tag{2.31}$$

2.2 Electric Potential

2.2.1 Electric Potential

The implication that curl of an electric field is zero allows us to infer using Stokes' Theorem that the closed path integral should also be zero.

$$\nabla \times \boldsymbol{E} = 0 \implies \oint_{\mathcal{P}} \boldsymbol{E} \cdot d\boldsymbol{\ell} = 0 \tag{2.32}$$

This is a property of what is called a *conservative field*, and from the Gradient theorem, we know that the electric field must be a gradient of a scalar function. This function is called the *electric potential*.

$$\boldsymbol{E} = -\nabla\varphi \tag{2.33}$$

Now, addition of a constant value to the potential will change it, but not the electric field. By convention, we take a reference point $\mathcal{O}$ at which we force the potential to be zero, which is at infinity. If we have some issues with the infinity, one can simply choose a different reference point.
Now, the electric potential of field is given by the following equation:

$$\varphi(\boldsymbol{r}) \equiv -\int_{\mathcal{O}}^{\boldsymbol{r}} \boldsymbol{E} \cdot d\boldsymbol{\ell} \tag{2.34}$$

and the potential difference between two points in space is:

$$\varphi(\boldsymbol{b}) - \varphi(\boldsymbol{a}) = -\int_{\boldsymbol{a}}^{\boldsymbol{b}} \boldsymbol{E} \cdot d\boldsymbol{\ell} \tag{2.35}$$

2.2.2 Poisson's and Laplace's Equation

Let's talk about Gauss' Law again. It basically just says that charges produce electric fields.

$$\nabla \cdot \boldsymbol{E} = \frac{\rho}{\varepsilon_0} \tag{2.36}$$

With our new knowledge of the scalar electric potential as $\boldsymbol{E} = -\nabla\varphi$, we obtain a new equation for our potential.

$$\nabla^2\varphi = -\frac{\rho}{\varepsilon_0} \tag{2.37}$$

This is called the *Poisson's Equation*. If there is no charge, then we obtain *Laplace's Equation*.

$$\nabla^2\varphi = 0 \tag{2.38}$$

There are special methods we use to solve these equations, that we shall explore in the future.
This is all a consequence of the curl of $\boldsymbol{E}$ being zero, it allows us to reduce our problem from solving two equations – Gauss' and Curl Law – into just needing to solve Poisson's or Laplace's equation.

2.2.3 Potential of Localized Distribution

Suppose we have a point charge q, for which we want to know the electric potential of, at a distance r away from the point. Using Equation (2.34), we write:

$$\varphi(r) = -\frac{1}{4\pi\varepsilon_0}\int_{\infty}^{r}\frac{q}{(r')^2}\,dr' \tag{2.39}$$

$$= \frac{1}{4\pi\varepsilon_0}\frac{q}{r} \tag{2.40}$$

In general, we write:

$$\varphi(\boldsymbol{r}) = \frac{1}{4\pi\varepsilon_0}\frac{q}{\boldsymbol{\imath}} \tag{2.41}$$

Electric potential follows the law of superposition (check it!), so the potential due to multiple point charges at a point is:

$$\varphi(\boldsymbol{r}) = \frac{1}{4\pi\varepsilon}\frac{q_i}{\boldsymbol{\imath}_i} \tag{2.42}$$

and for a continuous distribution:

$$\varphi(\boldsymbol{r}) = \frac{1}{4\pi\varepsilon_0}\iiint\frac{\rho(\boldsymbol{r}')}{\boldsymbol{\imath}}\,d\tau' \tag{2.43}$$

One must remember that these formulas only apply for a reference point chosen to $\mathcal{O} = \infty$. You cannot apply these formulas to charges that extend to infinity, the integral will diverge.

2.2.4 Field Discontinuity

Consider a charged sheet with uniform charge density σ and surface $\mathcal{S}$. Drawing a gaussian pillbox with both sides a very small distance ε away from the sheet, and applying Gauss' Law, we get:

$$\oiint_{\mathcal{S}}\boldsymbol{E}\cdot d\boldsymbol{a} = \frac{Q}{\varepsilon_0} = \frac{\sigma A}{\varepsilon_0} \tag{2.44}$$

$$E^{\perp}_{\text{above}} - E^{\perp}_{\text{below}} = \frac{\sigma}{\varepsilon_0} \tag{2.45}$$

Even in the limit that $\varepsilon \to 0$, we would have the difference in electric fields right below and above the surface to be σ/ε_0. This allows to conclude that when passing through a charged surface, the electric field experiences a discontinuity of σ/ε_0. Note that, the component parallel to the surface is unaffected. In general, we have the following:

$$\boldsymbol{E}_{\text{above}} - \boldsymbol{E}_{\text{below}} = \frac{\sigma}{\varepsilon_0}\hat{\boldsymbol{n}} \tag{2.46}$$

where $\hat{\boldsymbol{n}}$ is the unit vector perpendicular to the surface at that point. The potential meanwhile, is continuous across a boundary.

$$\varphi_{\text{above}} - \varphi_{\text{below}} = -\int_{\boldsymbol{a}}^{\boldsymbol{b}}\boldsymbol{E}\cdot d\boldsymbol{\ell} \tag{2.47}$$

As the path length shrinks down, $\boldsymbol{a} = \boldsymbol{b}$, and the integral is zero and hence $\varphi_{\text{below}} = \varphi_{\text{above}}$. However, the gradient of the potential is discontinuous, so we have:

$$\nabla\varphi_{\text{above}} - \nabla\varphi_{\text{below}} = -\frac{\sigma}{\varepsilon_0}\hat{\boldsymbol{n}} \tag{2.48}$$

$$\frac{\partial\varphi_{\text{above}}}{\partial n} - \frac{\partial\varphi_{\text{below}}}{\partial n} = -\frac{\sigma}{\varepsilon_0} \tag{2.49}$$

where $\partial\varphi/\partial n$ denotes the *normal derivative*, defined as:

$$\frac{\partial\varphi}{\partial n} = \nabla\varphi \cdot \hat{\boldsymbol{n}} \tag{2.50}$$

2.3 Electrostatic Work and Energy

2.3.1 Work to move Charges

The work done it takes to move a charge Q from points $\boldsymbol{a}$ to $\boldsymbol{b}$ through a path $\mathcal{P}$ is:

$$W = -\int_{\boldsymbol{a}}^{\boldsymbol{b}} \boldsymbol{F} \cdot d\boldsymbol{\ell} = -Q\int_{\boldsymbol{a}}^{\boldsymbol{b}} \boldsymbol{E} \cdot d\boldsymbol{\ell} = Q[\varphi(\boldsymbol{b}) - \varphi(\boldsymbol{a})] \tag{2.51}$$

Notice that the work done is independent of the path, and hence the electric force is conservative.

2.3.2 Energy of a Point Distribution

The work taken to assemble a collection of charges is the energy of the system. Say we bring in the first charge q_1, this takes no work since there is no other charge to influence it. Now, bringing in q_2 will cost you an energy of $q_2\varphi(\boldsymbol{r}_1)$, and say we put it at $\boldsymbol{r}_2$, then the work done would be:

$$W_2 = \frac{1}{4\pi\varepsilon_0} q_2 \left(\frac{q_1}{\imath_{12}}\right) \tag{2.52}$$

Bringing in q_3, you will have to the potentials of q_1 and q_2 to oppose you, and now you get:

$$W_3 = \frac{1}{4\pi\varepsilon_0} q_3 \left(\frac{q_1}{\imath_{13}} + \frac{q_2}{\imath_{23}}\right) \tag{2.53}$$

Similarly, bringing q_4 will take the work:

$$W_4 = \frac{1}{4\pi\varepsilon_0} q_4 \left(\frac{q_1}{\imath_{14}} + \frac{q_2}{\imath_{24}} + \frac{q_3}{\imath_{34}}\right) \tag{2.54}$$

The total work taken to assemble i.e. the energy of these four charges is hence:

$$W = \frac{1}{4\pi\varepsilon_0} \left(\frac{q_1q_2}{\imath_{12}} + \frac{q_1q_3}{\imath_{13}} + \frac{q_1q_4}{\imath_{14}} + \frac{q_2q_3}{\imath_{23}} + \frac{q_2q_4}{\imath_{24}} + \frac{q_3q_4}{\imath_{34}}\right) \tag{2.55}$$

The general rule for n charges is given as:

$$W = \frac{1}{8\pi\varepsilon_0}\frac{q_i q_j}{r_{ij}} \quad (j \neq i) \tag{2.56}$$

Note the use of Einstein summation convention. We can also write this in terms of potential as:

$$W = \frac{1}{2}q_i\left(\frac{1}{4\pi\varepsilon_0}\frac{q_j}{r_{ij}}\right) = \frac{1}{2}q_i\varphi_i(\boldsymbol{r}_i) \tag{2.57}$$

2.3.3 Energy of a Continuous Distribution

For charges distributed over a volume $\mathcal{V}$, the equation becomes:

$$W = \frac{1}{2}\iiint \rho\varphi \, d\tau \tag{2.58}$$

Using Gauss' Law, we rewrite $\rho = \varepsilon_0 \nabla \cdot \boldsymbol{E}$, then

$$W = \frac{1}{2}\varepsilon_0 \iiint (\nabla \cdot \boldsymbol{E})\varphi \, d\tau \tag{2.59}$$

Now, if one notices that $\nabla \cdot (\varphi \boldsymbol{E}) = \nabla\varphi \cdot \boldsymbol{E} + \varphi \nabla \cdot \boldsymbol{E}$, then we rewrite this as:

$$W = \frac{1}{2}\varepsilon_0 \left(\iiint \nabla \cdot (\varphi \boldsymbol{E}) \, d\tau - \iiint \nabla\varphi \cdot \boldsymbol{E} \, d\tau\right) \tag{2.60}$$

And now we use divergence theorem, and the fact that $-\nabla\varphi = \boldsymbol{E}$ to get:

$$W = \frac{1}{2}\varepsilon_0 \left(\oiint_{\mathcal{S}} \varphi \boldsymbol{E} \cdot d\boldsymbol{a} + \iiint_{\mathcal{V}} E^2 \, d\tau\right) \tag{2.61}$$

Now, if we let the volume considered be all of space, then the left integral becomes zero since we have no electric field or potential at the surface at infinity, since $E \propto 1/r^2$. Therefore, the electrostatic energy stored in all of space is given by:

$$W = \frac{1}{2}\varepsilon_0 \iiint E^2 \, d\tau \tag{2.62}$$

and the electrostatic energy density is given by:

$$u = \frac{1}{2}\varepsilon_0 E^2 \tag{2.63}$$

One must note that the energy does not follow the superposition principle, as one can check by replacing E with $E_1 + E_2$, you will not get $W_1 + W_2$.

2.4 Conductors

2.4.1 Properties of Conductors

In a metallic conductor, there are free electrons which move around the conductor when an potential is supplied to them. For an ideal conductor, we have the following properties:

1. $\boldsymbol{E} = 0$ inside the conductor. This is because if there were any field, then the charges in the conductor would move around due to the influence of that field and into the lowest energy configuration, that of no field.
2. $\rho = 0$ inside the conductor. This is just a consequence of Gauss' Law $\nabla \cdot \boldsymbol{E} = \rho/\varepsilon_0$, if $\boldsymbol{E} = 0$ then $\rho = 0$.
3. The surface charge density σ may or may not be zero, it is the only place where charges are allowed in a conductor.
4. A conductor is an equipotential body, this is once again a consequence of $\boldsymbol{E} = 0 \implies \Delta\varphi = -\int \boldsymbol{E} \cdot d\boldsymbol{\ell} = 0$.
5. $\boldsymbol{E}$ is perpendicular to the surface just outside the conductor. If not, then there would be a component inside the conductor, and the free electrons would rearrange and cancel it out.

2.4.2 Faraday Cage

Consider a spherical conductor, which has a cavity inside it. Let there be an electric field present outside the conductor. Now, the question is to find the electric field inside the cavity. First, we know the curl of the electric field is zero, then by Stokes' Theorem, the closed path integral is also zero.

$$\nabla \times \boldsymbol{E} = 0 \implies \oint \boldsymbol{E} \cdot d\boldsymbol{\ell} = 0 \tag{2.64}$$

Consider an arbitrary looped path $\mathcal{P}$ through the conductor, with a portion of the path $\mathcal{P}_1$ inside the cavity and the rest $\mathcal{P}_2$ inside the conductor. Since the we have zero electric field inside the conductor, that path integral falls to zero. Hence, we get:

$$\oint_{\mathcal{P}_1} \boldsymbol{E} \cdot d\boldsymbol{\ell} = 0 \tag{2.65}$$

Since $\mathcal{P}$ is arbitrary, then we conclude $\boldsymbol{E} = 0$. This property of conductors, when they enclose a volume, that volume will have zero electric field, is called a *Faraday cage*.

2.4.3 Force on a Conductor

Take a conductor, and lets zoom in far enough that we consider the surface to be flat. Using Gauss' Law, by taking a gaussian pillbox, we find the electric field immediately

outside the surface to be:

$$\oiint \boldsymbol{E} \cdot d\boldsymbol{a} = \frac{q}{\varepsilon_0} \tag{2.66}$$

$$\implies EA = \frac{\sigma A}{\varepsilon_0} \tag{2.67}$$

$$\implies \boldsymbol{E} = \frac{\sigma}{\varepsilon_0}\hat{\boldsymbol{n}} \tag{2.68}$$

We don't get $\sigma/2\varepsilon_0$ since the electric field inside the conductor is zero. Now, the equation we obtain for the electric field is consistent with our discontinuity condition,

$$\frac{\partial \varphi}{\partial n} = -\frac{\sigma}{\varepsilon_0} \tag{2.69}$$

Our goal is to calculate the force per unit area on the conductor. To do this, we simply take $\boldsymbol{f} = \sigma \boldsymbol{E}$, but what electric field do we consider? Say we have an electric field $\boldsymbol{E}$ on a patch of surface, then this is composed of the field due to the patch and field due to other sources.

$$\boldsymbol{E} = \boldsymbol{E}_{\text{patch}} + \boldsymbol{E}_{\text{other}} \tag{2.70}$$

Force cannot be due to $\boldsymbol{E}_{\text{patch}}$, since it is like trying to lift yourself by pushing on yourself, it doesn't work. The force is then due to other sources, and this does not face a discontinuity unlike the field due to the patch. Since above and below, we have a field due to the surface charge itself, we get:

$$\boldsymbol{E}_{\text{above}} = \boldsymbol{E}_{\text{other}} + \frac{\sigma}{2\varepsilon_0}\hat{\boldsymbol{n}} \tag{2.71}$$

$$\boldsymbol{E}_{\text{below}} = \boldsymbol{E}_{\text{other}} - \frac{\sigma}{2\varepsilon_0}\hat{\boldsymbol{n}} \tag{2.72}$$

$$\boldsymbol{E} = \boldsymbol{E}_{\text{other}} = \frac{1}{2}\left(\boldsymbol{E}_{\text{above}} + \boldsymbol{E}_{\text{below}}\right) \tag{2.73}$$

And so, for a conductor, the average electric field is $E = \sigma/2\varepsilon_0$ and we get the force per unit area as:

$$\boldsymbol{f} = \frac{\sigma^2}{2\varepsilon_0}\hat{\boldsymbol{n}} \tag{2.74}$$

This amounts to an outward electrostatic pressure, ignoring the sign of σ. Expressing it in terms of the electric field:

$$P = \frac{1}{2}\varepsilon_0 E^2 \tag{2.75}$$

2.5 Solving for Potential

This section will discuss various techniques one may require when solving differential or integral equations for the electric potential.

2.5.1 Earnshaw's Theorem

Statement: The scalar potential $\varphi(\boldsymbol{r})$ in a finite, charge-free region of space $\mathcal{V}$ takes its maximum or minimum values on the boundary of $\mathcal{V}$.

Proof: Suppose we have a point $\boldsymbol{r}_0$ which is a local minima inside the region $\mathcal{V}$. Let the surface of a small region enclosing $\boldsymbol{r}_0$ be $\mathcal{S}$, then for every point on this surface $\partial\varphi/\partial n = \hat{\boldsymbol{n}} \cdot \nabla\varphi > 0$, and it follows that:

$$\oiint_{\mathcal{S}} \nabla\varphi \cdot d\boldsymbol{a} > 0 \tag{2.76}$$

$$\implies \oiint_{\mathcal{S}} \boldsymbol{E} \cdot d\boldsymbol{a} < 0 \tag{2.77}$$

But by Gauss Law, this integral should be zero, hence we have a contradiction to our assumption that the local minima lies on an interior point. Using a similar argument, we can do this for local maxima as well. Hence, the solutions to Laplace's equation take extrema only at the boundaries. □

Earnshaw's theorem is also sometimes stated as *a charged particle cannot be held in by electrostatic forces alone*. This is a consequence of what we just proved, for there to be no force on a charge, the charge should sit at a point of local minima of potential, but that is only possible at a boundary point, where it cannot sit. Hence, charged particles cannot stay in an equilibrum just by electrostatic force.

2.5.2 Laplace's Equation

We obtained Poisson's Equation to be the following:

$$\nabla^2\varphi = -\frac{\rho}{\varepsilon_0} \tag{2.78}$$

In the case of zero charge density, this reduces to Laplace's Equation:

$$\nabla^2\varphi = 0 \tag{2.79}$$

Let's examine the nature of solutions to Laplace's equation in increasing dimensions.

One Dimension

Laplace's equation in one dimension is:

$$\frac{\partial^2\varphi}{\partial x^2} = 0 \tag{2.80}$$

The solution to this is obvious, it is a linear solution.

$$\varphi(x) = mx + c \tag{2.81}$$

Now, we explore the properties of this solution, so that we may find more general properties in higher dimensions. One thing we notices is that the potential at a point is the average of potential at points equidistant from that point, or mathematically,

$$\varphi(x) = \frac{\varphi(x+a) + \varphi(x-a)}{2} \tag{2.82}$$

The second property we notice is that extreme values i.e. minima or maxima of $\varphi(x)$ only occur at the boundaries. This is averaging is just a consequence of Earnshaw's Theorem.

Two Dimensions

Now, if we include a second dimension, Laplace's equation becomes:

$$\frac{\partial^2 \varphi}{\partial x^2} + \frac{\partial^2 \varphi}{\partial y^2} = 0 \tag{2.83}$$

Now, we can no longer solve this for a general solution, the solution will depend upon boundary conditions imposed. However, some properties still hold in higher dimensions. Earnshaw's theorem still holds, and the potential at a point is still the average of potential at points equidistant from it. In this case, we have to consider a circle $\mathcal{C}$ of radius R.

$$\varphi(x, y) = \frac{1}{2\pi R} \oint_{\mathcal{C}} \varphi \, d\ell \tag{2.84}$$

Three Dimensions

Laplace's equation in three dimensions becomes:

$$\frac{\partial^2 \varphi}{\partial x^2} + \frac{\partial^2 \varphi}{\partial y^2} + \frac{\partial^2 \varphi}{\partial z^2} = 0 \tag{2.85}$$

Similar to the two and one dimensional cases, we average here over a sphere's surface $\mathcal{S}$ of potential as a consequence of Earnshaw's Theorem.

$$\varphi(\boldsymbol{r}) = \frac{1}{4\pi R^2} \oiint_{\mathcal{S}} \varphi \, da \tag{2.86}$$

2.5.3 Uniqueness Theorems

First Uniqueness Theorem

Statement: The solution to Laplace's equation in some volume $\mathcal{V}$ is unique if φ is specified on its boundary surface $\mathcal{S}$.

Proof: Let φ_1 and φ_2 be two solutions to Laplace's equation with our boundary condition in that volume $\mathcal{V}$ withs surface $\mathcal{S}$.

$$\nabla^2\varphi_1 = 0 \quad \nabla^2\varphi_2 = 0$$

Now, we take $\varphi_3 \equiv \varphi_2 - \varphi_1$, and this is also a solution to Laplace's equation.

$$\nabla^2\varphi_3 = \nabla^2\varphi_2 - \nabla^2\varphi_1 = 0$$

But φ_3 takes value zero at the boundaries, since φ_1 and φ_2 are equal there, and by Earnshaw's theorem, the minimum and maximum value of φ_3 is zero and hence $\varphi_3 = 0$. Therefore, $\varphi_1 = \varphi_2$, and the solution with this boundary condition is unique. □

Second Uniqueness Theorem

Statement: In a volume $\mathcal{V}$ surrounded by conductors with a given charge density ρ, the electric field is uniquely determined if the total charge on each conductor is given.

Proof: Suppose we have two fields $\boldsymbol{E}_1$ and $\boldsymbol{E}_2$ that satisfy Gauss' law in the space between conductors.

$$\nabla \cdot \boldsymbol{E}_1 = \nabla \cdot \boldsymbol{E}_2 = \frac{\rho}{\varepsilon_0}$$

and in integral form, for a surface enclosing each conductor, we have:

$$\oiint_{\mathcal{S}_i} \boldsymbol{E}_1 \cdot d\boldsymbol{a} = \oiint_{\mathcal{S}_i} \boldsymbol{E}_2 \cdot d\boldsymbol{a} = \frac{Q_i}{\varepsilon_0}$$

Likewise, for the outer boundary, we have:

$$\oiint_{\mathcal{S}} \boldsymbol{E}_1 \cdot d\boldsymbol{a} = \oiint_{\mathcal{S}} \boldsymbol{E}_2 \cdot d\boldsymbol{a} = \frac{Q_i}{\varepsilon_0}$$

Examining the difference, we get:

$$\boldsymbol{E}_3 \equiv \boldsymbol{E}_1 - \boldsymbol{E}_2 \implies \nabla \cdot \boldsymbol{E}_3 = 0$$

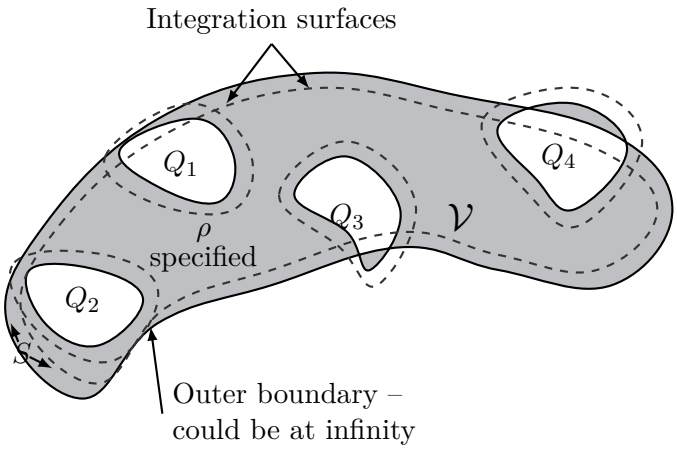
This is in the region between the conductors, and over each boundary surface, we get:

$$\oiint \boldsymbol{E}_3 \cdot d\boldsymbol{a} = 0$$

Although we do not know how the charge Q_i is distributed over each surface, we do know that conductors have equipotential surfaces. If we let $\varphi_3 = \varphi_1 - \varphi_2$, then

$$\nabla \cdot (\varphi_3 \boldsymbol{E}_3) = \varphi_3(\nabla \cdot \boldsymbol{E}_3) + \boldsymbol{E}_3 \cdot (\nabla\varphi_3) = -E_3^2$$

Integrating this over the entire region $\mathcal{V}$, and using divergence theorem,



$$\iiint_P = \mathcal{V}\nabla \cdot (\varphi_3 \boldsymbol{E}_3)\, d\tau = \oiint_{\mathcal{S}} \varphi_3 \boldsymbol{E}_3 \cdot d\boldsymbol{a} = -\iiint_{\mathcal{V}} (E_3)^2\, d\tau$$

Since φ_3 is constant over each surface if we set the boundary to be infinity, and what remains is $\oiint \boldsymbol{E}_3 \cdot d\boldsymbol{a}$ which is we showed is zero. Hence,

$$\iiint (E_3)^2\, d\tau = 0 \implies E_3 = 0 \implies \boldsymbol{E}_1 = \boldsymbol{E}_2$$

Hence, we obtain that the solution to these boundary conditions is unique. □

2.5.4 Method of Images

Consider the following system: You have a infinitely large conducting plate with its surfac being the x-y plane, and this conductor is grounded so the potential on the surface is zero. You also have a point charge q at a height $z = d$ from the surface. Now, our problem is to find the electric potential for this system and the charge density induced on the surface plate. Essentially, we have to solve Poisson's equation:

$$\nabla^2 \varphi = -\frac{\rho}{\varepsilon_0}$$

This situation ends up being pretty complicated to solve, but lets start by finding the boundary conditions here:

1. $\varphi = 0$ when $z = 0$
2. $\varphi \to 0$ for $\imath \to \infty$

A corollary of the first uniqueness theorem can be applied here, and we can say that this case has a unique solution. So if we can find a different setup that satisfies these boundary conditions, we have solved the problem.

Analyse the given setup: You have two charges q and $-q$ at distances $z = d$ and $z = -d$ from the x-y plane respectively. The potential generated by this setup of charges is:

$$\varphi(x, y, z) = \frac{q}{4\pi\varepsilon_0}\left[\frac{1}{\sqrt{x^2 + y^2 + (z-d)^2}} - \frac{1}{\sqrt{x^2 + y^2 + (z+d)^2}}\right] \tag{2.87}$$

Notice that this completely solves our boundary conditions! Hence, this the potential that solves our initial problem for $z \geq 0$. We will refer to this trick system that we consider as the image system.
Now, we use this to find the surface charge induced on the plate. From (2.49), and using the fact that the potential under the plate is zero, we write:

$$\frac{\partial\varphi}{\partial n} = -\frac{\sigma}{\varepsilon_0} \tag{2.88}$$

Here, the normal direction is the $\hat{\boldsymbol{z}}$ direction, hence our equation becomes:

$$\sigma = -\varepsilon_0 \frac{\partial\varphi}{\partial z}\Big|_{z=0} \tag{2.89}$$

And we obtain the surface charge density to be:

$$\sigma(x, y) = \frac{-qd}{2\pi(x^2 + y^2 + d^2)^{3/2}} \tag{2.90}$$

The energy stored in this system should be calculated only considering half the space and only one charge, one should not get carried away and assume everything is the same between the two systems. The only thing we need from the image system is the potential, and that is all. In this case, it is easier to get the potential energy by calculating the work needed to bring the charge from infinity to $(0, 0, d)$. Note that the force is the same in both systems, since its just a consequence of the same potential, just that you are working on one charge here and 2 charges in the image system. There it would take twice as much work.

$$\boldsymbol{F} = -\frac{1}{4\pi\varepsilon_0}\frac{q^2}{4z^2}\hat{\boldsymbol{z}} \tag{2.91}$$

$$V = \int \boldsymbol{F}\cdot d\boldsymbol{\ell} = \frac{1}{4\pi\varepsilon_0}\int_\infty^d \frac{q^2}{4z^2}\,dz = -\frac{1}{4\pi\varepsilon_0}\frac{q^2}{4d} \tag{2.92}$$

2.5.5 Multipole Expansion

We wish to approximate the potential of a system by a series expansion. To do this, consider the potential of a localized charge distribution.

$$\varphi(\boldsymbol{r}) = \frac{1}{4\pi\varepsilon_0}\iiint \frac{\rho(\boldsymbol{r}')}{\boldsymbol{z}}\,d\tau' \tag{2.93}$$

Let the angle between $\boldsymbol{r}$ and $\boldsymbol{r}'$ be α, then we write:

$$\begin{aligned} \boldsymbol{\imath} &= |\boldsymbol{r} - \boldsymbol{r}'| = \sqrt{r^2 + (r')^2 - 2rr'\cos\alpha} && (2.94)\\ &= r\sqrt{1 + \left(\frac{r'}{r}\right)^2 - 2\left(\frac{r'}{r}\right)\cos\alpha} && (2.95)\\ &= r\sqrt{1+\varepsilon} && (2.96)\\ & && (2.97) \end{aligned}$$

where

$$\varepsilon \equiv \left(\frac{r'}{r}\right)\left(\frac{r'}{r} - 2\cos\alpha\right) \tag{2.98}$$

Given that $r' \ll r$, using the binomial expansion, we write:

$$\frac{1}{\boldsymbol{\imath}} = \frac{1}{r}(1+\varepsilon)^{-1/2} = \frac{1}{2}\left(1 - \frac{1}{2}\varepsilon + \frac{3}{8}\varepsilon^2 - \frac{5}{16}\varepsilon^3 + \dots\right) \tag{2.99}$$

which simplifies to:

$$\frac{1}{\boldsymbol{\imath}} = \frac{1}{r}\sum_{n=0}^{\infty}\left(\frac{r'}{r}\right)^n P_n(\cos\alpha) \tag{2.100}$$

where $P_n(x)$ are the *Legendre polynomials*, the first few polynomials given below:

$$\begin{aligned} P_0(x) &= 1 && (2.101)\\ P_1(x) &= x && (2.102)\\ P_2(x) &= \frac{1}{2}(3x^2 - 1) && (2.103)\\ P_3(x) &= \frac{1}{2}(5x^3 - 3x) && (2.104)\\ &\vdots && (2.105)\\ P_n(x) &= \frac{1}{2^n n!}\frac{d^n}{dx^n}(x^2-1)^n && (2.106) \end{aligned}$$

Therefore, the electric potential becomes:

$$\varphi(\boldsymbol{r}) = \frac{1}{4\pi\varepsilon_0}\sum_{n=0}^{\infty}\frac{1}{r^{n+1}}\iiint (r')^n P_n(\cos\alpha)\rho(\boldsymbol{r}')\,d\tau' \tag{2.107}$$

Monopoles and Dipoles

The series is dominated by the monopole terms, which is:

$$\varphi_{\text{mon}}(\boldsymbol{r}) = \frac{1}{4\pi\varepsilon_0}\frac{Q}{r} \tag{2.108}$$

where $Q = \iiint \rho \, d\tau$ is the total charge. If there is no total charge, then potential comes majorly from the dipole term,

$$\varphi_{\text{di}}(\boldsymbol{r}) = \frac{1}{4\pi\varepsilon_0}\frac{1}{r^2}\iiint r' \cos\alpha \, \rho(\boldsymbol{r}') \, d\tau' \tag{2.109}$$

Since α is the angle between $\boldsymbol{r}$ and $\boldsymbol{r}'$, then $r' \cos\alpha = \hat{\boldsymbol{r}} \cdot \boldsymbol{r}'$, and hence:

$$\varphi_{\text{di}}(\boldsymbol{r}) = \frac{1}{4\pi\varepsilon_0}\frac{1}{r^2}\hat{\boldsymbol{r}} \cdot \iiint \boldsymbol{r}' \rho(\boldsymbol{r}') \, d\tau' \tag{2.110}$$

and we define the *dipole moment* of the distribution:

$$\boldsymbol{p} = \iiint \boldsymbol{r}' \rho(\boldsymbol{r}') \, d\tau' \tag{2.111}$$

Hence, the potential of the dipole simplifies to:

$$\varphi_{\text{di}}(\boldsymbol{r}) = \frac{1}{4\pi\varepsilon_0}\frac{\boldsymbol{p} \cdot \hat{\boldsymbol{r}}}{r^2} \tag{2.112}$$

For a discrete distribution of charges, the dipole moment is given by:

$$\boldsymbol{p} = q_i \boldsymbol{r}_i' \tag{2.113}$$

One should note the importance of where the origin is placed when analyzing a multipole. If a single point charge is at the origin, then that is a monopole, otherwise its is not and you will have a dipole term as well.
Consider a dipole, with dipole moment $\boldsymbol{p}$ originally. Now, shift the origin to a point $\boldsymbol{a}$, then let the new dipole moment be $\bar{\boldsymbol{p}}$, then:

$$\bar{\boldsymbol{p}} = \iiint \bar{\boldsymbol{r}}' \rho(\boldsymbol{r}') \, d\tau' = \iiint (\boldsymbol{r}' - a) \rho(\boldsymbol{r}') \, d\tau' \tag{2.114}$$

$$= \iiint \boldsymbol{r}' \rho(\boldsymbol{r}') \, d\tau' - \boldsymbol{a} \iiint \rho(\boldsymbol{r}') \, d\tau' = \boldsymbol{p} - Q\boldsymbol{a} \tag{2.115}$$

Now, we wish to find the electric field of a dipole. Let's choose a coordinate system such that $\boldsymbol{p}$ lies at the origin and points in the $\hat{\boldsymbol{z}}$ direction, then the potential at (r, θ) will be:

$$\varphi_{\text{di}}(\boldsymbol{r}) = \frac{1}{4\pi\varepsilon_0}\frac{\hat{\boldsymbol{r}} \cdot \boldsymbol{p}}{r^2} = \frac{p\cos\theta}{4\pi\varepsilon_0 r^2} \tag{2.116}$$

Taking the gradient, we get the electric field to be:

$$\boldsymbol{E}_{\text{di}}(r, \theta) = \frac{p}{4\pi\varepsilon_0 r^3}(2\cos\theta \, \hat{\boldsymbol{r}} + \sin\theta \, \hat{\boldsymbol{\theta}}) \tag{2.117}$$

Using $\boldsymbol{p} = p \, \hat{\boldsymbol{z}}$ and $\hat{\boldsymbol{z}} = \cos\theta \, \hat{\boldsymbol{r}} - \sin\theta \, \hat{\boldsymbol{\theta}}$, we write that:

$$\boldsymbol{E}_{\text{di}}(r, \theta) = \frac{1}{4\pi\varepsilon_0 r^3}\left[3(\boldsymbol{p} \cdot \hat{\boldsymbol{r}})\hat{\boldsymbol{r}} - \boldsymbol{p}\right] \tag{2.118}$$

Chapter 3

Electric Fields in Matter

"As our ultimate physical sources of field are not, we believe, infinite concentrations of charge in zero volume but instead finite structures, we simply ignore the mathematical singularities implied by our point-charge language and rule out of bounds the interior of our elementary sources."

Electricity and Magnetism, 3rd Ed p. 2021
Edward M. Purcell

3.1 Polarization

We shall begin a study of dielectrics – which is the second broad class of materials, after conductors. In conductors, the charges are free to move about the volume, whereas in dielectrics, the electrons are tightly bound by the nucleus and stay fixed. Electric fields affect dielectrics by two mechanisms: stretching or rotating distributions in atoms/molecules.

3.1.1 Induced Dipoles

When a neutral atom is placed in the presence of an electric field $\boldsymbol{E}$, the nucleus and the electron cloud are affected by the field, with both drifting in either directions until a balance is reached. The atom now has a dipole and is said to be *polarized*. The dipole moment is in the same direction as the field, hence we have:

$$\boldsymbol{p} = \alpha \boldsymbol{E} \tag{3.1}$$

where α is said to be the *atomic polarizibility*. For molecules, we have a similar situation. However, due to the orientation and distribution of charges, the polarizibility is different

in each direction, hence it is described by a tensor.

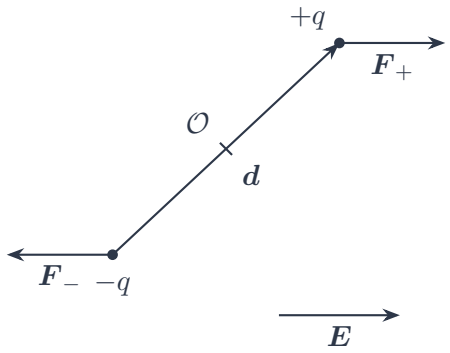
$$\begin{pmatrix} p_x \\ p_y \\ p_z \end{pmatrix} = \begin{pmatrix} \alpha_{xx} & \alpha_{xy} & \alpha_{xz} \\ \alpha_{yx} & \alpha_{yy} & \alpha_{yz} \\ \alpha_{zx} & \alpha_{zy} & \alpha_{zz} \end{pmatrix} \begin{pmatrix} E_x \\ E_y \\ E_z \end{pmatrix} \tag{3.2}$$

$$\boldsymbol{p} = \boldsymbol{\alpha} \boldsymbol{E} \tag{3.3}$$

Consider a polar molecule (molecule with a net dipole moment) placed in an electric field. If the dipole is given by $\boldsymbol{p} = q\boldsymbol{d}$, then we have a torque on either ends given by $\boldsymbol{F}_+ = q\boldsymbol{E}$ and $\boldsymbol{F}_- = -q\boldsymbol{E}$.

$$\boldsymbol{N} = (\boldsymbol{r}_+ \times \boldsymbol{F}_+) + (\boldsymbol{r}_- \times \boldsymbol{F}_-) \tag{3.4}$$

$$= q(\boldsymbol{d} \times \boldsymbol{E}) = \boldsymbol{p} \times \boldsymbol{E} \tag{3.5}$$



The total force on the dipole is given by:

$$\boldsymbol{F} = \boldsymbol{F}_+ \boldsymbol{F}_- = q(\boldsymbol{E}_+ - \boldsymbol{E}_-) = q(\Delta \boldsymbol{E}) \tag{3.6}$$

and we can approximate $\Delta \boldsymbol{E}$ by $\Delta E_i = (\nabla E_i) \cdot \boldsymbol{d}$, or as $\Delta \boldsymbol{E} = (\boldsymbol{d} \cdot \nabla)\boldsymbol{E}$. Hence, we have:

$$\boldsymbol{F} = (\boldsymbol{p} \cdot \nabla)\boldsymbol{E} \tag{3.7}$$

And about any point $\boldsymbol{r}$, the torque is given by:

$$\boldsymbol{F} = (\boldsymbol{p} \times \boldsymbol{E}) + (\boldsymbol{r} \times \boldsymbol{F}) \tag{3.8}$$

3.1.2 Bound Charge

We proceed by defining a new quantity: the *polarization* $\boldsymbol{P}$ of a dielectric, which is the dipole moment per volume of dielectric.

$$\boldsymbol{p} = \iiint \boldsymbol{P}\, d\tau \tag{3.9}$$

Now, writing out the expression for the potential due to a dipole,

$$\varphi(\boldsymbol{r}) = \frac{1}{4\pi\varepsilon_0}\frac{\hat{\boldsymbol{\imath}}\cdot\boldsymbol{p}}{\imath^2} \tag{3.10}$$

and we replace the dipole moment with the polarization:

$$\varphi(\boldsymbol{r}) = \frac{1}{4\pi\varepsilon_0}\iiint \frac{\hat{\boldsymbol{\imath}}\cdot\boldsymbol{P}(\boldsymbol{r}')}{\imath^2}\,d\tau' \tag{3.11}$$

Observe that $\nabla'(1/\imath) = \hat{\boldsymbol{\imath}}/\imath^2$, then:

$$\varphi(\boldsymbol{r}) = \frac{1}{4\pi\varepsilon_0}\iiint \boldsymbol{P}\cdot\nabla'\left(\frac{1}{\imath}\right)\,d\tau' \tag{3.12}$$

and using divergence product rules, we get:

$$\varphi(\boldsymbol{r}) = \frac{1}{4\pi\varepsilon_0}\left[\iiint \nabla'\cdot\left(\frac{\boldsymbol{P}}{\imath}\right)\,d\tau' - \iiint \frac{1}{\imath}(\nabla'\cdot\boldsymbol{P})\,d\tau'\right] \tag{3.13}$$

and now using Divergence theorem, we obtain:

$$\varphi(\boldsymbol{r}) = \frac{1}{4\pi\varepsilon_0}\left[\oiint \frac{1}{\imath}\boldsymbol{P}\cdot d\boldsymbol{a}' - \iiint \frac{1}{\imath}(\nabla'\cdot\boldsymbol{P})\,d\tau'\right] \tag{3.14}$$

The first term looks a surface charge density and the second a volume charge density, so we define the surface and volume bound charge densities as:

$$\sigma_b = \boldsymbol{P}\cdot\hat{\boldsymbol{n}} \tag{3.15}$$

$$\rho_B = -\nabla\cdot\boldsymbol{P} \tag{3.16}$$

With these new definitions, we have the potential described by:

$$\varphi(\boldsymbol{r}) = \frac{1}{4\pi\varepsilon_0}\oiint_{\mathcal{S}}\frac{\sigma_b}{\imath}\,da' + \iiint_{\mathcal{V}}\frac{\rho_b}{\imath}\,d\tau' \tag{3.17}$$

It is left to the reader to check that:

$$\oiint_{\mathcal{S}}\sigma_b\,da + \iiint_{\mathcal{V}}\rho_b\,d\tau = 0 \tag{3.18}$$

The bound charge represents the charges trapped by an atom/molecule in a dielectric. They are not just mathematically convenient definitions, they really do exist. The charge not bound is called free charge.

3.2 Electric Displacement

3.2.1 Gauss' Law in Dielectrics

We found the bound charge densities to be $\sigma_b = \boldsymbol{P} \cdot \hat{\boldsymbol{n}}$ and $\rho_b = -\nabla \cdot \boldsymbol{P}$. The total charge density will be the sum $\rho = \rho_f + \rho_b$, where ρ_f the free charge density. Using Gauss' law,

$$\varepsilon_0 \nabla \cdot \boldsymbol{E} = \rho = \rho_f + \rho_b \tag{3.19}$$

$$= \rho_f - \nabla \cdot \boldsymbol{P} \tag{3.20}$$

$$\nabla \cdot (\varepsilon_0 \boldsymbol{E} + \boldsymbol{P}) = \rho_f \tag{3.21}$$

and we define the *Electric Displacement* vector as follows:

$$\boldsymbol{D} = \varepsilon_0 \boldsymbol{E} + \boldsymbol{P} \tag{3.22}$$

then Gauss' Law reads:

$$\nabla \cdot \boldsymbol{D} = \rho_f \tag{3.23}$$

or in integral form,

$$\oiint \boldsymbol{D} \cdot d\boldsymbol{a} = Q_f \tag{3.24}$$

Note that there does not exist a Coulomb's law for the electric displacement, since that requires the curl of $\boldsymbol{D}$ to be zero.

$$\boldsymbol{D}(\boldsymbol{r}) \neq \frac{1}{4\pi} \iiint \frac{\hat{\boldsymbol{\imath}}}{\imath^2} \rho(\boldsymbol{r}')\, d\tau' \tag{3.25}$$